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62.mlx

61.mlx

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41.mlx

61.asv

61.mlx

62.mlx

Addition and multiplication....

causal non causal 42.mlx

continuous time system.mlx

Discrete.mlx

Workspace

Name	Value	Size	Class
t	1×501 double	1×501	double
t_conv	1×1001 dou...	1×1001	double
x1	1×501 double	1×501	double
x2	1×501 double	1×501	double
y	1×1001 dou...	1×1001	double

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clc;

clear;

close all;

% Time axis

t = 0:0.01:5;

% Input signals

x1 = exp(-t); % Exponential signal

x2 = ones(size(t)); % Unit step signal

% Convolution

y = conv(x1, x2) * 0.01;

% Time axis for output

t_conv = 0:0.01:(length(y)-1)*0.01;

Signal $x_1(t)=e^{-t}$

Amplitude

Time (s)

Signal $x_2(t)=u(t)$

Amplitude

Time (s)

Convolution Output $y(t)$

Amplitude

Time (s)

Command Window

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```

% Convolution of Discrete-Time Signals
clc;
clear;
close all;
% Input sequences
x1 = [1 2 3 4];
x2 = [1 1 1];
% Convolution
y = conv(x1,x2);
% Sample indices
n1 = 0:length(x1)-1;
n2 = 0:length(x2)-1;
ny = 0:length(y)-1;
% Display output
disp('Convolution Output:');
disp(y);

```

Convolution Output:

1 3 6 9 7 4

Amplitude

Input Sequence $x_1(n)$

0 1 2 3

n

Amplitude

Input Sequence $x_2(n)$

-0.5 0 0.5 1 1.5 2 2.5

n

Amplitude

Convolution Output $y(n)$

0 1 2 3 4 5

n

Command Window

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